

ADSS Practical Assignment 1

User Instruction Manual

A practical guide to the menu options, data entry, reports, and pricing logic.

Purpose

This program manages products, category chains, discounts, stock levels, faulty product reports, and inventory summaries.

Prepared from the program menu and feature descriptions provided by the user.

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1. Program Overview

This console-based inventory management program helps users create and organize products, manage category structures, update stock information, apply discounts, report faulty products, and generate useful inventory reports.

Best practice

Before adding products, first create the category structure. Each product must belong to a complete category chain: main category, subcategory, and sub-subcategory.

2. Main Menu

Option	Action
1	Add Product
2	Add Category / Subcategory / Sub-subcategory
3	Add Faulty Product (Report Damage)
4	Update Product
5	Add Product / Category Discount
6	View Stock Alerts
7	Export Faulty Inventory Report by Dates
8	Add Supplier Discount per Product

Option	Action
9	Get Inventory Report by Categories
10	Show All Products
11	Purchase Product from Shelves / Stock
12	Find Product Price for Consumer
13	Create Test Data
0	Exit

3. Feature Guide

Option 1 — Add Product

Use this feature to create a new product record.

The user will be prompted to enter or select:

- Product name
- Catalog number
- Main category
- Subcategory
- Sub-subcategory
- Storage location
- Manufacturer
- Amount on shelves
- Amount in stock
- Minimum amount alert
- Supply price
- Consumer price

A product can be created only if at least one full category chain already exists: main category → subcategory → sub-subcategory.

Option 2 — Add Category / Subcategory / Sub-subcategory

This feature opens a submenu for creating category levels.

1. Create a main category

The user enters the category name and a discount percentage (a double between 0 and 1).

2. Create a subcategory

The user enters the category name and a discount percentage, then selects the main category to which the new subcategory will be attached.

3. Create a sub-subcategory

The user enters the category name and a discount percentage, then selects both the main category and the subcategory to which the new sub-subcategory will be attached.

Option 3 — Add Faulty Product (Report Damage)

Use this feature to report a faulty or damaged product.

The user will be prompted for:

- Catalog number
- Current location
- Description of the fault

The date and time of the report are saved automatically in the system.

Option 4 — Update Product

Use this feature to update an existing product.

A menu is displayed, and the user selects which field to update:

- a. Name (string)
- b. Storage location (string)
- c. Consumer price (double)
- d. Supplier price (double)
- e. Amount on shelves (int)
- f. Amount in stock (int)
- g. Minimum amount alert (int)

After selecting a field, the user enters the new value.

Option 5 — Add Product / Category Discount

Use this feature to define a discount for either a category or a single product.

1. Category discount

The user selects one main category and enters a new discount percentage (a double between 0 and 1).

2. Product discount

The user enters the product's catalog number and a discount percentage.

Option 6 — View Stock Alerts

This feature displays all products that are in the run-out range.

A product p is considered in the run-out range if:

$p.\text{amountOnShelves} + p.\text{amountOnStock} \leq p.\text{minAmountAlert}$

Option 7 — Export Faulty Inventory Report by Dates

This feature shows all faulty product reports within a selected date range.

The user is prompted for:

- Start date
- End date

The date format is dd/mm/yyyy.

The system then displays all faulty reports created within that range.

Option 8 — Add Supplier Discount per Product

Use this feature to set a supplier discount for a specific product.

The user will be prompted for:

- Catalog number
- Discount percentage (double between 0 and 1)

Option 9 — Get Inventory Report by Categories

This feature generates an inventory report for selected main categories.

For each included product, the report shows:

- Product name
- Catalog number
- Location
- Amount on shelves
- Amount in stock

The user selects a main category from the list, and after each selection can choose:

- y — add another category
- n — finish the request

Option 10 — Show All Products

This feature displays all products stored in the system.

For each product, the program shows every available detail, including:

- Product name
- Catalog number
- Category chain
- Storage location
- Manufacturer
- Amount on shelves

- Amount in stock
- Minimum amount alert
- Supply price
- Consumer price
- Supplier discount percentage
- Product discount percentage

Option 11 — Purchase Product from Shelves / Stock

This feature simulates purchasing a product and is mainly intended for testing.

The user is prompted for:

- Catalog number
- Quantity to buy from shelves
- Quantity to buy from stock

Negative numbers may also be entered. In that case, the values in the system will increase rather than decrease.

Option 12 — Find Product Price for Consumer

This feature calculates the final consumer price of a product, including discounts.

The user is prompted for the product's catalog number.

The final price is calculated as:

$(\text{base consumer price}) \times (1 - \text{product discount}) \times (1 - \text{category discount})$

Option 13 — Create Test Data

This feature creates a small fixed data set for testing purposes.

4. Product Pricing and Discounts

The program supports several discount-related actions. Product discounts affect a single product, while category discounts apply to products through their category structure. Supplier discounts are stored separately from the final price shown to the consumer.

Final price formula

$\text{Final consumer price} = (\text{base consumer price}) \times (1 - \text{product discount}) \times (1 - \text{category discount})$

5. Data Entry Rules and Notes

- Discount percentages are entered as doubles between 0 and 1. For example, 0.15 represents a 15% discount.
- Dates for faulty inventory reports are entered in dd/mm/yyyy format.

- Stock alerts are triggered when the sum of shelf quantity and stock quantity is less than or equal to the minimum amount alert.
- Negative quantities in the purchase feature can be used to increase inventory values during testing.